

OWNER'S MANUAL

Operation
Maintenance
Specifications

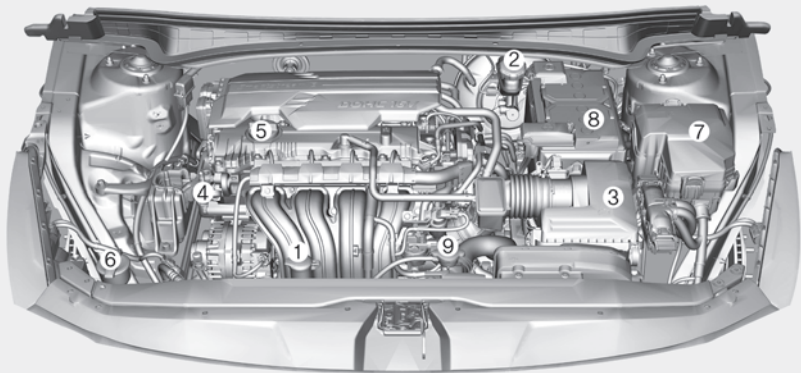
All information in this Owner's Manual is current at the time of publication. However, HYUNDAI reserves the right to make changes at any time so that our policy of continual product improvement may be carried out.

This manual applies to all models of this vehicle and includes descriptions and explanations of optional as well as standard equipment.

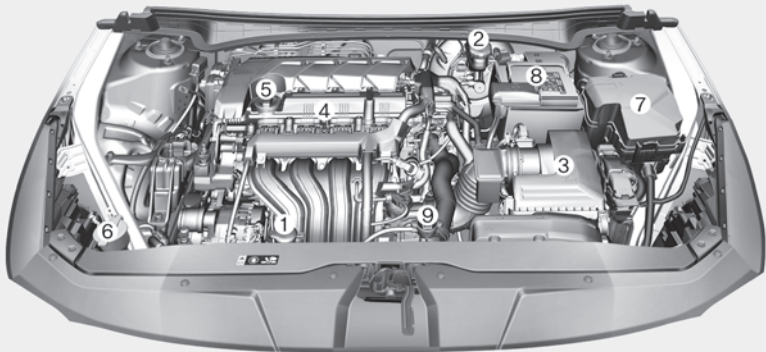
As a result, you may find material in this manual that does not apply to your specific vehicle.

ENGINE COMPARTMENT

■ Smartstream G2.0 MPI



■ Smartstream G1.6



The actual engine compartment in the vehicle may differ from the illustration.

OCN7080064L/OCN7080001

1. Engine coolant reservoir	9-25	5. Engine oil filler cap	9-23
2. Brake/clutch fluid reservoir	9-28	6. Windshield washer fluid reservoir	9-29
3. Air cleaner	9-30	7. Fuse box	9-48
4. Engine oil dipstick	9-23	8. Battery	9-35
		9. Radiator cap	9-23

TIRES AND WHEELS

Items	Tire size	Wheel size	Inflation pressure bar (kPa, psi)				Wheel bolt torque kgf-m (lbf-ft, N-m)
			Normal load *1		Maximum load		
			Front	Rear	Front	Rear	
Full size tire	195/65 R15	6.0Jx15	2.35 (235, 34)	2.15 (215, 31)	2.4 (240, 35)	2.4 (240, 35)	11~13 (79~94, 107~127)
	205/55 R16	6.5Jx16	2.35 (235, 34)	2.15 (215, 31)	2.4 (240, 35)	2.4 (240, 35)	
	225/45 R17	7.0Jx17	2.35 (235, 34)	2.15 (215, 31)	2.4 (240, 35)	2.4 (240, 35)	
Compact spare tire (if equipped)	T125/80 D15	4Tx15	4.2 (420, 60)	4.2 (420, 60)	4.2 (420, 60)	4.2 (420, 60)	

*1: Normal load : Up to 3 persons

NOTICE

- It is permissible to add 20 kPa (3 psi) to the standard tire pressure specification if colder temperatures are expected soon. Tires typically lose 7 kPa (1 psi) for every 7°C (12°F) temperature drop. If extreme temperature variations are expected, recheck your tire pressure as necessary to keep them properly inflated.
- An air pressure generally decreases, as you drive up to a high-altitude area above sea level. Thus, if you plan to drive a high-altitude area, check the tire pressures in advance. If necessary, inflate them to a proper level (Air inflation per altitude: +10 kPa/1 km (+2.4 psi/1 mile)).



CAUTION

When replacing tires, use the same size originally supplied with the vehicle. Using tires of a different size can damage the related parts or not work properly

RECOMMENDED LUBRICANTS AND CAPACITIES

To help achieve proper engine and powertrain performance and durability, use only lubricants of the proper quality. The correct lubricants also help promote engine efficiency that results in improved fuel economy.

These lubricants and fluids are recommended for use in your vehicle.

Lubricant		Volume	Classification	
Engine oil *1 (drain and refill) Recommends 	Smartstream G2.0 MPI	4.3 ℓ (4.54 US qt.)	Except Middle East *3	0W-20 API SN PLUS/SP or ILSAC GF-6 *1
			For Middle East *3	5W-30(ACEA A5/B5) or 0W-30(API SN PLUS/SP or ILSAC GF-6) *1
	Smartstream G1.6	3.8 ℓ (4.02 US qt.)	Except Middle East *3	SAE 0W-20, API SN PLUS/SP or ILSAC GF-6 *2
			For Middle East *3	SAE 5W-30, API Latest(SN PLUS) or ACEA A5/B5 *2
	1.6 D-CVVT	3.6 ℓ (3.80 US qt.)	Except Middle East *3	SAE 5W-20 / API Latest (ILSAC Latest) *4 *5
			For Middle East *3	SAE 5W-30 / ACEA A5/B5 *4 *5
Manual transmission fluid	Smartstream G2.0 MPI	1.5~1.6 ℓ (1.6~1.7 US qt.)	HK SYN MTF 70W (SK) SPIRAX S6 GHME 70W MTF (H.K.SHELL) GS MTF HD 70W (GS CALTEX) (API GL-4, SAE 70W, TGO-9)	
	Smartstream G1.6	1.6~1.7 ℓ (1.7~1.8 US qt.)		
	1.6 D-CVVT	1.6~1.7 ℓ (1.7~1.8 US qt.)		
Automatic transmission fluid		7.5 ℓ (7.9 US qt.)	MICHANG ATF SP-IV, SK ATF SP-IV, NOCA ATF SP-IV, HYUNDAI genuine ATF SP-IV	
Intelligent variable transmission fluid		6.5 ℓ (6.87 US qt.)	Hyundai genuine SP-CVT1	
Coolant	Smartstream G2.0 MPI	7.4 ℓ (7.82 US qt.)	Mixture of antifreeze and water (Phosphate-based Ethylene glycol coolant for aluminum radiator)	
	Smartstream G1.6	6.1 ℓ (6.45 US qt.)		
	1.6 D-CVVT	7.1 ℓ (7.5 US qt.)		
Brake/clutch fluid		0.7~0.8 ℓ (0.74~0.85 US qt.)	DOT-4 **	
Fuel		47 ℓ (49.66 US qt.)	Refer to “Fuel requirements” in the Foreword chapter.	

- *1: If the above recommended specification oil is not available, SAE 0W-20 grade synthetic oil can be used. If mineral oil or semi-synthetic oil is used, it is a severe maintenance condition in terms of engine oil change.
- *2: If the above recommended specification oil is not available, SAE 0W-30/5W-30 grade synthetic oil can be used. If mineral oil or semi-synthetic oil is used, it is a severe condition in terms of engine oil change.
- *3: Middle East includes Libya, Algeria, Morocco, Egypt, Tunisia, Sudan, Iran, etc.
- *4: If the recommended engine oil is not available in your country, you are able to use API SL / ILSAC GF-3 or ACEA A3.
- *5: Refer to the recommended SAE viscosity numbers on page 14.
- *6: To maintain the best braking performance and ABS/ESC performance, we recommend that you use genuine brake fluid that conform to specifications. (Standard : SAE J1704 DOT-4 LV, ISO4925 CLASS-6 and FMVSS116 DOT-4

Recommended SAE viscosity number



CAUTION

Always be sure to clean the area around any filler plug, drain plug, or dipstick before checking or draining any lubricant. This is especially important in dusty or sandy areas and when the vehicle is used on unpaved roads. Cleaning the plug and dipstick areas will prevent dirt and grit from entering the engine and other mechanisms that could be damaged.

Engine oil viscosity (thickness) has an effect on fuel economy and cold weather operating (engine start and engine oil flowability). Lower viscosity engine oils can provide better fuel economy and cold weather performance, however, higher viscosity engine oils are required for satisfactory lubrication in hot weather. Using oils of any viscosity other than those recommended could result in engine damage.

When choosing an oil, consider the range of temperature your vehicle will be operated in before the next oil change.

Proceed to select the recommended oil viscosity from the chart.

Temperature Range for SAE Viscosity Numbers											
Temperature		°C	-30	-20	-10	0	10	20	30	40	50
		(°F)	-10	0	20	40	60	80	100	120	
Smartstream G2.0 MPI	For Middle East										
		0/5W-30									
	Except Middle East										
		0W-20									
Smartstream G1.6	For Middle East										
		0W-20									
	Except Middle East										
		5W-30									
1.6 D-CVVT	For Middle East										
						20W-50					
					15W-40						
				10W-30							
	5W-30, 5W-40										
	Except Middle East										
0W-40, 5W-20, 0/5W-30, 5W-40											

BRAKING SYSTEM

Power brakes

Your vehicle has power-assisted brakes that adjust automatically through normal usage.

If the engine is not running or is turned off while driving, the power assist for the brakes will not work. You can still stop your vehicle by applying greater force to the brake pedal than typical. The stopping distance, however, will be longer than with power brakes.

When the engine is not running, the reserve brake power is partially depleted each time the brake pedal is applied. Do not pump the brake pedal when the power assist has been interrupted.

Pump the brake pedal only when necessary to maintain steering control on slippery surfaces.



WARNING

Take the following precautions:

- **Do not drive with your foot resting on the brake pedal. This will create abnormal high brake temperatures, excessive brake lining and pad wear, and increased stopping distances.**
- **When descending down a long or steep hill, downshift to a lower gear in order to control your speed without using the brake pedal excessively. Applying the brakes continuously will cause the brakes to overheat and could result in a temporary loss of braking performance.**

- **Wet brakes may impair the vehicle's ability to safely slow down; the vehicle may also pull to one side when the brakes are applied. Applying the brakes lightly will indicate whether they have been affected in this way. Always test your brakes in this fashion after driving through deep water. To dry the brakes, lightly tap the brake pedal to heat up the brakes while maintaining a safe forward speed until brake performance returns to normal. Avoid driving at high speeds until the brakes function correctly.**

Disc brakes wear indicator

When your brake pads are worn and new pads are required, you will hear a high pitched warning sound from your front or rear brakes. You may hear this sound come and go or it may occur whenever you depress the brake pedal.

Note that some driving conditions or climates may cause a brake squeal when you first apply (or lightly apply) the brakes. This is normal and does not indicate a problem with your brakes.

NOTICE

To avoid costly brake repairs, do not continue to drive with worn brake pads.



Information

Always replace brake pads as complete front or rear axle sets.

9. Maintenance

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SCHEDULED MAINTENANCE SERVICES

Follow Normal Maintenance Schedule if the vehicle is usually operated where none of the following conditions apply.

If any of the following conditions apply, you must follow the Maintenance Under Severe Usage Conditions.

- Repeated driving short distance of less than 8 km (5 miles) in normal temperature or less than 16 km(10 miles) in freezing temperature
- Low speed driving for long distances
- Driving on rough, dusty, muddy, unpaved, graveled or salt-spread roads
- Driving in areas using salt or other corrosive materials or in very cold weather
- Driving in heavy dust condition
- Driving in heavy traffic area
- Driving on uphill, downhill, or mountain road repeatedly
- Towing a trailer or using a camper, or roof rack
- Driving as a patrol car, taxi, other commercial use of vehicle towing
- Driving over 170 km/h (106 mile/h)
- Frequently driving in stop-and-go condition
- Engine oil usage which not recommended(mineral, semi-synthetic, viscosity grade)

If your vehicle is operated under the above conditions, you should inspect, replace or refill more frequently than the following Normal Maintenance Schedule. After the periods or distance shown in the chart, continue to follow the prescribed maintenance intervals.

Normal Maintenance Schedule – Gasoline Engine (Except Europe)

MAINTENANCE ITEM	MAINTENANCE INTERVALS	Number of months or driving distance, whichever comes first										
		Months	12	24	36	48	60	72	84	96		
		Miles×1,000	10	20	30	40	50	60	70	80		
		Km×1,000	15	30	45	60	75	90	105	120		
Drive belts *1				I		I		I		I		I
Engine oil and engine oil filter *2		For Middle East *3, Central & South America	Replace every 10,000 km (6,200 miles) or 12 months									
		Except Middle East *3, Central & South America	R	R	R	R	R	R	R	R		R
Air cleaner filter		For Middle East *3	R	R	R	R	R	R	R	R		R
		Except Middle East *3	I	I	R	I	I	R	I	I		I
Fuel additives *4			Add every 10,000 km (6,200 miles) or 12 months									
Spark plugs *5	2.0 MPI	Unleaded fuel	Replace every 160,000 km (100,000 miles)									
		Leaded fuel	Replace every 80,000 km (50,000 miles)									
	Smartstream G1.6	Unleaded fuel	Replace every 160,000 km (100,000 miles)									
		Unleaded fuel	Replace every 60,000 km (40,000 miles)									
	1.6 D-CVT	Unleaded fuel	Replace every 30,000 km (20,000 miles)									
		Leaded fuel	Replace every 30,000 km (20,000 miles)									

I : Inspect and if necessary, adjust, correct, clean or replace.

R : Replace or change.

*1: Adjust alternator and water pump drive belt air conditioner drive belt (if equipped). Inspect and if necessary correct or replace.

*2: Check the engine oil level and leak every 500 km (350 miles) or before starting a long trip.

*3: Middle East includes Morocco, Sudan and Egypt.

*4: If good quality gasolines meet Europe Fuel standards (EN228) or equivalents including fuel additives is not available, one bottle of additive is recommended. Additives are available from your authorized HYUNDAI dealer along with information on how to use them. Do not mix other additives.

*5: For your convenience, it can be replaced prior to it's interval when you do maintenance of other items.

Normal Maintenance Schedule (Cont.) – Gasoline Engine (Except Europe)

MAINTENANCE ITEM	MAINTENANCE INTERVALS	Number of months or driving distance, whichever comes first											
		Months	12	24	36	48	60	72	84	96			
		Miles×1,000	10	20	30	40	50	60	70	80			
		Km×1,000	15	30	45	60	75	90	105	120			
Valve clearance *6	1.6 D-CVVT							I					
Vapor hose and fuel filler cap						I				I			
Fuel tank air filter (if equipped)				I		R		I		R			
Vacuum hose			I	I	I	I	I	I	I	I			
Fuel filter *7				I		R		I		R			
Fuel lines, hoses and connections						I				I			
Cooling system			Inspect "Coolant level adjustment and leak" every day										
			At first, inspect 60,000 km (40,000 miles) or 48 months after that, inspect every 30,000 km (20,000 miles) or 24 months										

I : Inspect and if necessary, adjust, correct, clean or replace.

R : Replace or change.

*6 : Inspect for excessive valve noise and/or engine vibration and adjust if necessary. We recommend that the system be checked by an authorized HYUNDAI dealer.

*7 : The fuel filter is considered to be maintenance free but periodic inspection is recommended for this maintenance schedule depends on fuel quality. If there are some important matters like fuel flow restriction, surging, loss of power, hard starting problem etc, we recommend replacing the fuel filter immediately regardless of maintenance schedule and consulting with an authorized HYUNDAI dealer for details.

Normal Maintenance Schedule (Cont.) – Gasoline Engine (Except Europe)

MAINTENANCE ITEM	MAINTENANCE INTERVALS		Number of months or driving distance, whichever comes first									
	Months		12	24	36	48	60	72	84	96		
	Miles×1,000		10	20	30	40	50	60	70	80		
	Km×1,000		15	30	45	60	75	90	105	120		
Engine coolant *8			At first replace at 200,000 km (120,000 miles) or 10 years: after that, replace every 40,000 km (30,000 miles) or 24 months *9									
Battery condition		For Middle East	Inspect every 10,000 km (6,200 miles) or 6 months									
		Except Middle East	I	I	I	I	I	I	I	I	I	I
Brake lines, hoses and connections			I	I	I	I	I	I	I	I	I	I
Brake pedal, Clutch pedal (if equipped)				I		I		I		I		I
Parking brake				I		I		I		I		I
Brake/clutch fluid			I	I	R	I	I	R	I	I	I	I
Disc brakes and pads			I	I	I	I	I	I	I	I	I	I
Steering gear rack, linkage and boots			I	I	I	I	I	I	I	I	I	I

I : Inspect and if necessary, adjust, correct, clean or replace.

R : Replace or change.

*8 : When adding coolant, use only deionized water or soft water for your vehicle and never mix hard water in the coolant filled at the factory. An improper coolant mixture can result in serious malfunction or engine damage.

*9 : For your convenience, it can be replaced prior to it's interval when you do maintenance of other items.

Normal Maintenance Schedule (Cont.) – Gasoline Engine (Except Europe)

MAINTENANCE ITEM	MAINTENANCE INTERVALS	Number of months or driving distance, whichever comes first											
		Months	12	24	36	48	60	72	84	96			
		Miles×1,000	10	20	30	40	50	60	70	80			
		Km×1,000	15	30	45	60	75	90	105	120			
Driveshaft and boots				I		I		I		I			I
Tire (pressure & tread wear)			I	I	I	I	I	I	I	I			I
Front suspension ball joints			I	I	I	I	I	I	I	I			I
Bolt and nuts on chassis and body			I	I	I	I	I	I	I	I			I
Air conditioner refrigerant (if equipped)			I	I	I	I	I	I	I	I			I
Air conditioner compressor (if equipped)			I	I	I	I	I	I	I	I			I
Climate control air filter (if equipped)			R	R	R	R	R	R	R	R			R
Manual transmission fluid (if equipped) *10						I							I
Automatic transmission fluid (if equipped)			No check, No service required										
Intelligent variable transmission (if equipped)			No check, No service required										
Exhaust system				I		I		I		I			I

I : Inspect and if necessary, adjust, correct, clean or replace.

R : Replace or change.

*10 : Manual transmission/Dual clutch transmission fluid should be changed anytime they have been submerged in water.

Maintenance Under Severe Usage and Low Mileage Conditions – Gasoline Engine (Except Europe)

The following items must be serviced more frequently on cars mainly used under severe driving conditions. Refer to the chart below for the appropriate maintenance intervals.

R : Replace

I : Inspect and if necessary, adjust, correct, clean or replace

Maintenance item		Maintenance operation	Maintenance Intervals	Driving condition
Engine oil and engine oil filter	For Middle East, Central & South America	R	Replace every 5,000 km (3,100 miles) or 6 months	A, B, C, D, E, F, G, H, I, K, L
	Except Middle East, Central & South America	R	Replace every 7,500 km (4,650 miles) or 6 months	
Air cleaner filter		R	Replace more frequently depending on the condition	C, E
Spark plugs		R	Replace more frequently depending on the condition	A, B, H, I, K
Steering gear rack, linkage and boots		I	Inspect more frequently depending on the condition	C, D, E, F, G
Front suspension ball joints		I	Inspect more frequently depending on the condition	C, D, E, F, G

Maintenance item	Maintenance operation	Maintenance Intervals	Driving condition
Disc brakes and pads, calipers and rotors	I	Inspect more frequently depending on the condition	C, D, E, G, H
Parking brake	I	Inspect more frequently depending on the condition	C, D, G, H
Driveshaft and boots	I	Inspect more frequently depending on the condition	C, D, E, F, G, H, I
Manual transmission fluid (if equipped)	R	Every 120,000 km (80,000 miles)	C, D, E, F, G, H, I, J
Automatic transmission fluid (if equipped)	R	Every 90,000 km (60,000 miles)	A, C, D, E, F, G, H, I, K
Intelligent variable transmission (if equipped)	R	Every 90,000 km (60,000 miles)	A, C, D, E, F, G, H, I, K
Climate control air filter (if equipped)	R	Replace more frequently depending on the condition	C, E

Severe driving conditions

- A. Repeatedly driving short distance of less than 8 km (5 miles) in normal temperature or less than 16 km (10 miles) in freezing temperature
- B. Low speed driving for long distances
- C. Driving on rough, dusty, muddy, unpaved, graveled or salt spread roads
- D. Driving in areas using salt or other corrosive materials or in very cold weather
- E. Driving in heavy dust condition
- F. Driving in heavy traffic area
- G. Driving on uphill, downhill, or mountain roads repeatedly
- H. Towing a trailer, or using a camper or roof rack
- I. Driving for patrol car, taxi, commercial car or vehicle towing
- J. Driving over 170 km/h (106 mile/h)
- K. Frequently driving in stop-and-go conditions
- L. Engine oil usage which not recommended(mineral, semi-synthetic, viscosity grade)

EXPLANATION OF SCHEDULED MAINTENANCE ITEMS

Engine oil and filter

The engine oil and filter should be changed at the intervals specified in the maintenance schedule. If the vehicle is being driven in severe conditions, more frequent oil and filter changes are required.

Drive belts

Inspect all drive belts for evidence of cuts, cracks, excessive wear or oil saturation and replace if necessary. Drive belts should be checked periodically for proper tension and adjusted as necessary.



CAUTION

When you are inspecting the belt, place the ignition switch to the lock/off or ACC position.

Fuel filter (Except Europe)

A clogged filter can limit the speed at which the vehicle may be driven, damage the emission system and cause multiple issues such as hard starting.

If an excessive amount of foreign matter accumulates in the fuel tank, the filter may require replacement more frequently.

After installing a new filter, run the engine for several minutes, and check for leaks at the connections. We recommend that the fuel filter be replaced by an authorized HYUNDAI dealer.

Fuel filter (For Europe)

This gasoline powered vehicle is equipped with a lifetime fuel filter that is integrated with the fuel tank. Regular maintenance or replacement is not needed; however, the quality of fuel used may impact the frequency of maintenance needed. If there are any fuel related problems like fuel flow restriction, surging, loss of power, hard starting problem etc, fuel filter inspection or replacement may be needed. We recommend the fuel filter be inspected or replaced by an authorized HYUNDAI dealer.

Fuel lines, fuel hoses and connections

Check the fuel lines, fuel hoses and connections for leakage and damage. We recommend that the fuel lines, fuel hoses and connections be replaced by an authorized HYUNDAI dealer.

Vapor hose and fuel filler cap

The vapor hose and fuel filler cap should be inspected at those intervals specified in the maintenance schedule. Make sure that a new vapor hose or fuel filler cap is correctly replaced.

Air cleaner filter

We recommend that the air cleaner filter be replaced by an authorized HYUNDAI dealer.

Spark plugs

Make sure to install new spark plugs of the correct heat range.



WARNING

Do not disconnect and inspect spark plugs when the engine is hot. You may burn yourself.

Valve clearance

Inspect excessive valve noise and/or engine vibration and adjust if necessary. We recommend that the system be serviced by an authorized HYUNDAI dealer.

Cooling system

Check the cooling system parts, such as radiator, coolant reservoir, hoses and connections for leakage and damage. Replace any damaged parts.

Engine coolant

The coolant should be changed at the intervals specified in the maintenance schedule.

Automatic transmission fluid (if equipped)

Automatic transmission fluid should not be checked under normal usage conditions.

We recommend that the automatic transmission fluid is changed by an authorized HYUNDAI dealer according to the maintenance schedule.



Information

Automatic transmission fluid color is red when new.

As the vehicle is driven, the automatic transmission fluid will begin to look darker.

This is a normal condition. It does not need to be replaced based on the color change.

NOTICE

The use of a non-specified fluid could result in transmission malfunction and failure.

Use only specified automatic transmission fluid. (Refer to “Recommended lubricants and capacities” in chapter 2.)

Manual transmission fluid (if equipped)

Inspect the manual transmission fluid according to the maintenance schedule.

Intelligent variable transmission fluid (if equipped)

Intelligent variable transmission fluid should not be checked under normal usage conditions.

We recommend that the Intelligent variable transmission fluid is changed by an authorized HYUNDAI dealer according to the maintenance schedule.

Brake hoses and lines

Visually check for proper installation, chafing, cracks, deterioration and any leakage. Replace any deteriorated or damaged parts immediately.

Brake/clutch fluid

Check the brake/clutch fluid level in the brake/clutch fluid reservoir. The level should be between “MIN” and “MAX” marks on the side of the reservoir. Use only hydraulic brake/clutch fluid conforming to DOT 4 specification.

Parking brake

Inspect the parking brake system.

Brake pads, calipers and rotors

Check the pads for excessive wear, discs for run out and wear, and calipers for fluid leakage.

For more information on checking the pads or lining wear limit, refer to the HYUNDAI web site.

(<http://service.hyundai-motor.com>)

Suspension mounting bolts

Check the suspension connections for looseness or damage. Retighten to the specified torque.

Steering gear box, linkage & boots/lower arm ball joint

With the vehicle stopped and engine off, check for excessive free-play in the steering wheel.

Check the linkage for bends or damage. Check the dust boots and ball joints for deterioration, cracks, or damage. Replace any damaged parts.

Drive shafts and boots

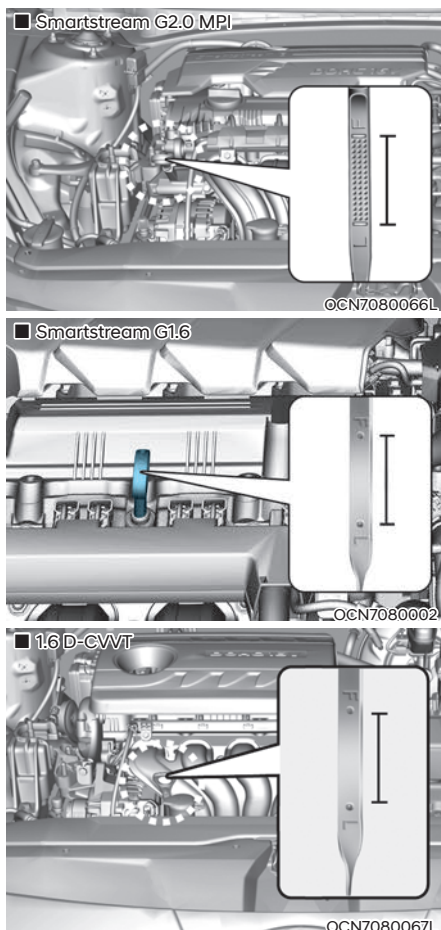
Check the drive shafts, boots and clamps for cracks, deterioration, or damage. Replace any damaged parts and, if necessary, repack the grease.

Air conditioning refrigerant/compressor

Check the air conditioning lines and connections for leakage and damage.

ENGINE OIL

Checking the engine oil level (Gasoline engine)



1. Be sure the vehicle is on level ground.
2. Start the engine and allow it to reach normal operating temperature.
3. Turn the engine off and wait for a few minutes (about 10 minutes) for the oil to return to the oil pan.
4. Pull the dipstick out, wipe it clean, and re-insert it fully.



WARNING

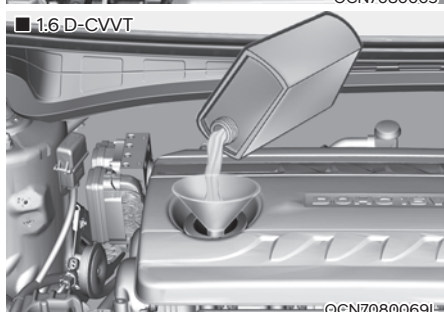
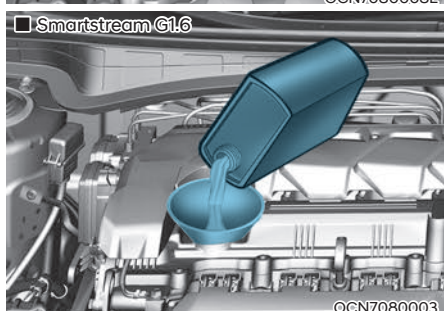
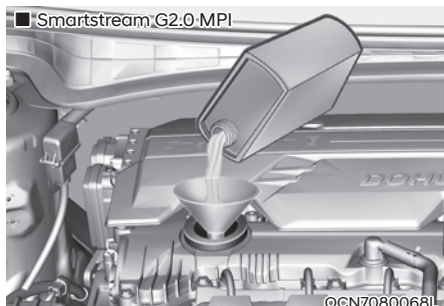
Radiator hose

Be very careful not to touch the radiator hose when checking or adding the engine oil as it may be hot enough to burn you.

5. Pull the dipstick out again and check the level. The level should be between F and L.

NOTICE

- Do not overfill the engine oil. It may damage the engine.
- Do not spill engine oil, when adding or changing engine oil. If you drop the engine oil on the engine room, wipe it off immediately.
- When you wipe the oil level gauge, you should wipe it with a clean cloth. When mixed with debris, it can cause engine damage.



If it is near or at L, add enough oil to bring the level to F. Do not overfill.

Use a funnel to help prevent oil from being spilled on engine components.

Use only the specified engine oil. (Refer to "Recommended lubricants and capacities" in chapter 2.)

Checking the engine oil and filter



We recommend that the engine oil and filter be replaced by an authorized HYUNDAI dealer.



WARNING

Used engine oil may cause irritation or cancer of the skin if left in contact with the skin for prolonged periods of time. Used engine oil contains chemicals that have caused cancer in laboratory animals. Always protect your skin by washing your hands thoroughly with soap and warm water as soon as possible after handling used oil.

ENGINE COOLANT

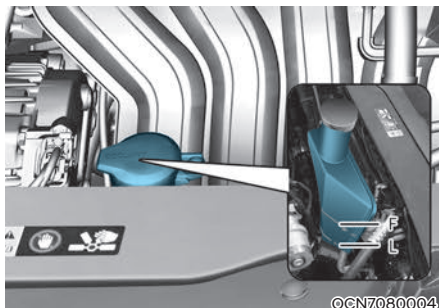
The high-pressure cooling system has a reservoir filled with year-round antifreeze coolant. The reservoir is filled at the factory.

Check the antifreeze protection and coolant concentration level at least once a year, at the beginning of the winter season, and before traveling to a colder climate.

NOTICE

- When the engine overheats from low engine coolant, suddenly adding engine coolant may cause cracks in the engine. To prevent damage, add engine coolant slowly in small quantities.
- Do not drive with no engine coolant. It may cause water pump failure and engine seizure, etc.

Checking the coolant level



WARNING



Removing radiator cap/ Engine coolant reservoir cap

- Never attempt to remove the radiator cap/engine coolant reservoir cap while the engine is operating or hot. Doing so might lead to cooling system and engine damage and could result in serious personal injury from escaping hot coolant or steam.
- Turn the engine off and wait until it cools down. Use extreme care when removing the radiator cap/engine coolant reservoir cap. Wrap a thick towel around it, and turn it counterclockwise slowly to the first stop. Step back while the pressure is released from the cooling system. When you are sure all the pressure has been released, press down on the cap, using a thick towel, and continue turning counterclockwise to remove it.

- **Even if the engine is not operating, do not remove the radiator cap/ engine coolant reservoir cap or the drain plug while the engine and radiator are hot. Hot coolant and steam may still blow out under pressure, causing serious injury.**



WARNING



The electric motor (cooling fan) is controlled by engine coolant temperature, refrigerant pressure and vehicle speed.

It may sometimes operate even when the engine is not running. Use extreme caution when working near the blades of the cooling fan so that you are not injured by a rotating fan blades. As the engine coolant temperature decreases, the electric motor will automatically shut off. This is a normal condition.

The electric motor (cooling fan) may operate until you disconnect the negative battery cable.

Check the condition and connections of all cooling system hoses and heater hoses. Replace any swollen or deteriorated hoses.

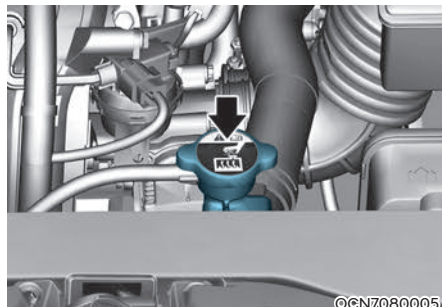
The coolant level should be filled between F (MAX) and L (MIN) marks on the side of the coolant reservoir when the engine is cool.

If the coolant level is low, add enough distilled (deionized) water. Bring the level to F (MAX), but do not overfill.

If frequent additions are required, we recommend that the system be inspected by an authorized HYUNDAI dealer.



WARNING



1. **Check if the radiator cap label is straight In front.**
2. **Make sure that the tiny protrusions inside the radiator cap should be securely interlocked.**

Recommended engine coolant

- When adding coolant, use only distilled (deionized) water for your vehicle and never mix hard water in the coolant filled at the factory. An incorrect coolant mixture can result in serious malfunction or engine damage.
- The engine in your vehicle has aluminum engine parts and must be protected by an phosphate based ethylene-glycol coolant to prevent corrosion and freezing.
- DO NOT USE alcohol or methanol coolant or mix them with the specified coolant.
- Do not use a solution that contains more than 60% antifreeze or less than 35% antifreeze, which would reduce the effectiveness of the solution.
- The engine in your vehicle has aluminum engine parts and must be protected by an ethylene-glycol with phosphate based coolant to prevent corrosion and freezing.

For mixture percentage, refer to the following table.

Ambient Temperature	Mixture Percentage (volume)	
	Antifreeze	Water
-15°C (5°F)	35	65
-25°C (-13°F)	40	60
-35°C (-31°F)	50	50
-45°C (-49°F)	60	40

Information

If in doubt about the mix ratio, a 50% water and 50% antifreeze mix is the easiest to mix together as it will be the same quantity of each. It is suitable to use for most temperature ranges of -35°C (-31°F) and higher.

Changing the engine coolant

We recommend that the coolant be replaced by an authorized HYUNDAI dealer.

NOTICE

Put a thick cloth or fabric around the radiator cap before refilling the coolant in order to prevent the coolant from overflowing into engine parts such as alternator.

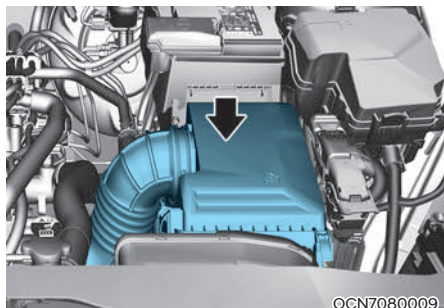
WARNING

Do not use engine coolant or antifreeze in the washer fluid reservoir.

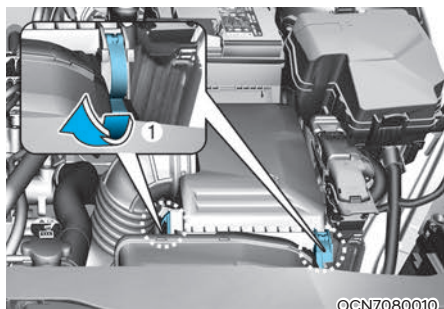
Engine coolant can severely obscure visibility when sprayed on the windshield and may cause loss of vehicle control or damage to paint and body trim.

AIR CLEANER

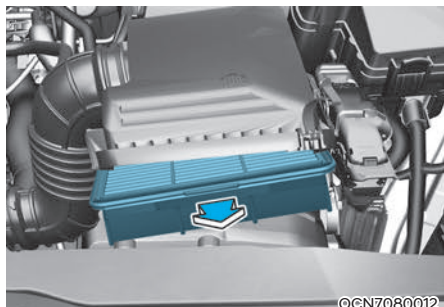
Filter replacement



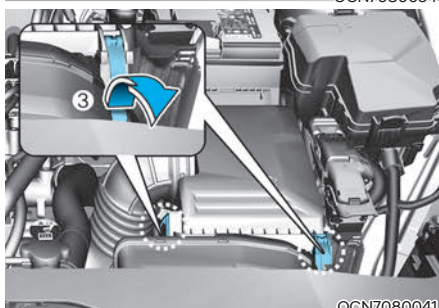
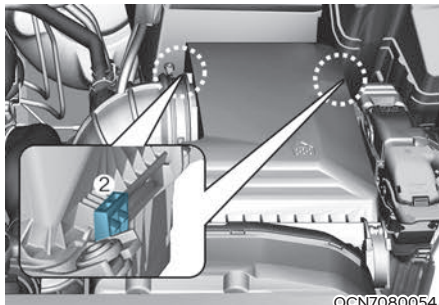
The air cleaner filter can be cleaned for inspection using compressed air. Do not attempt to wash or to rinse it, as water will damage the filter. If soiled, the air cleaner filter must be replaced.



1. Loosen the air cleaner cover attaching clip (1) and open the cover.



2. Replace the air cleaner filter.



3. Insert the air cleaner cover in the hinge (2) and engage the clip (3) after closing the cover.
4. Check that the cover is firmly installed.

i Information

If the vehicle is operated in extremely dusty or sandy areas, replace the element more often than the usual recommended intervals (refer to “Maintenance Under Severe Usage Conditions” in this chapter).

NOTICE

- Do not drive with the air cleaner filter removed. This will result in excessive engine wear.
- When removing the air cleaner filter, be careful that dust or dirt does not enter the air intake, or damage may result.
- Use HYUNDAI genuine parts or the equivalent specified for your vehicle. Use of parts without the matching quality could damage the air flow sensor.

TIRES AND WHEELS



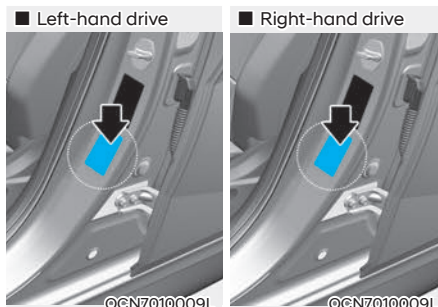
WARNING

Tire failure may cause loss of vehicle control resulting in an accident. To reduce risk of **SERIOUS INJURY** or **DEATH**, take the following precautions:

- Inspect your tires monthly for proper inflation as well as wear and damage.
- The recommended cold tire pressure for your vehicle can be found in this manual and on the tire label located on the driver's side center pillar. Always use a tire pressure gauge to measure tire pressure. Tires with too much or too little pressure wear unevenly causing poor handling.
- Check the pressure of the spare every time you check the pressure of the other tires on your vehicle.
- Replace tires that are worn, show uneven wear, or are damaged. Worn tires can cause loss of braking effectiveness, steering control, or traction.
- **ALWAYS** replace tires with the same size as each tire that was originally supplied with this vehicle. Using tires and wheels other than the recommended sizes could cause unusual handling characteristics, poor vehicle control, or negatively affect your vehicle's Anti-Lock Brake System (ABS) resulting in a serious accident.

Tire care

For proper maintenance, safety, and maximum fuel economy, you must always maintain recommended tire inflation pressures and stay within the load limits and weight distribution recommended for your vehicle.



All specifications (sizes and pressures) can be found on a label attached to the driver's side center pillar.

Recommended cold tire inflation pressures

All tire pressures (including the spare) should be checked when the tires are cold. "Cold tires" means the vehicle has not been driven for at least three hours or driven less than 1.6 km (one mile).

Warm tires normally exceed recommended cold tire pressures by 28 to 41 kPa (4 to 6 psi). Do not release air from warm tires to adjust the pressure or the tires will be under-inflated. For recommended inflation pressure, refer to "Tire and Wheels" in chapter 2.



WARNING

Recommended pressures must be maintained for the best ride, vehicle handling, and minimum tire wear.

Over-inflation or under-inflation can reduce tire life, adversely affect vehicle handling, and lead to sudden tire failure that could result in loss of vehicle control resulting in an accident.

Severe under-inflation can lead to severe heat build-up, causing blowouts, tread separation and other tire failures that can result in the loss of vehicle control resulting in an accident. This risk is much higher on hot days and when driving for long periods at high speeds.



CAUTION

- Underinflation also results in excessive wear, poor handling and reduced fuel economy. Wheel deformation also is possible. Keep your tire pressures at the proper levels. If a tire frequently needs refilling, we recommend that the system be checked by an authorized HYUNDAI dealer.
- Overinflation produces a harsh ride, excessive wear at the center of the tire tread, and a greater possibility of damage from road hazards.

Check tire inflation pressure

Check your tires, including the spare tire, once a month or more.

How to check

Use a good quality tire pressure gauge to check tire pressure. You can not tell if your tires are properly inflated simply by looking at them. Radial tires may look properly inflated when they are under-inflated.

Remove the valve cap from the tire valve stem. Press the tire gauge firmly onto the valve to get a pressure measurement. If the cold tire inflation pressure matches the recommended pressure on the tire and loading information label, no further adjustment is necessary. If the pressure is low, add air until you reach the recommended pressure. Make sure to put the valve caps back on the valve stems. Without the valve cap, dirt or moisture could get into the valve core and cause air leakage. If a valve cap is missing, install a new one as soon as possible.

If you overfill the tire, release air by pushing on the metal stem in the center of the tire valve. Recheck the tire pressure with the tire gauge. Be sure to put the valve caps back on the valve stems. Without the valve cap, dirt or moisture could get into the valve core and cause air leakage. If a valve cap is missing, install a new one as soon as possible.

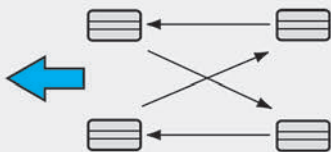
Tire rotation

To equalize tread wear, HYUNDAI recommends that the tires be rotated every 12,000 km (7,500 miles) or sooner if irregular wear develops.

During rotation, check the tires for correct balance.

When rotating tires, check for uneven wear and damage. Abnormal wear is usually caused by incorrect tire pressure, improper wheel alignment, out-of-balance wheels, severe braking or severe cornering. Look for bumps or bulges in the tread or side of the tire. Replace the tire if you find any of these conditions. Replace the tire if fabric or cord is visible. After rotation, be sure to bring the front and rear tire pressures to specification and check lug nut tightness (proper torque is 11~13 kgf·m [79~94 lbf·ft]).

■ Without a spare tire



OH1078078

Disc brake pads should be inspected for wear whenever tires are rotated.

i Information

The outside and inside of the unsymmetrical tire is distinguishable. When installing an unsymmetrical tire, be sure to install the side marked “outside” face the outside. If the side marked “inside” is installed on the outside, it will have a negative effect on vehicle performance.

! WARNING

- **Do not use the compact spare tire for tire rotation.**
- **Do not mix bias ply and radial ply tires under any circumstances. This may cause unusual handling characteristics that may cause loss of vehicle control resulting in an accident.**

Wheel alignment and tire balance

The wheels on your vehicle were aligned and balanced carefully at the factory to give you the longest tire life and best overall performance.

In most cases, you will not need to have your wheels aligned again. However, if you notice unusual tire wear or your vehicle pulling one way or the other, the alignment may need to be reset.

If you notice your vehicle vibrating when driving on a smooth road, your wheels may need to be rebalanced.

NOTICE

Incorrect wheel weights can damage your vehicle's aluminum wheels. Use only approved wheel weights.

Tire replacement



If the tire is worn evenly, a tread wear indicator will appear as a solid band across the tread. This shows there is less than 1.6 mm (1/16 in.) of tread left on the tire. Replace the tire when this happens.

Do not wait for the band to appear across the entire tread before replacing the tire.



WARNING

To reduce the risk of DEATH or SERIOUS INJURY:

- Replace tires that are worn, show uneven wear, or are damaged. Worn tires can cause loss of braking effectiveness, steering control, and traction.
- Always replace tires with the same size as each tire that was originally supplied with this vehicle. Using tires and wheels other than the recommended sizes could cause unusual handling characteristics, poor vehicle control, or negatively affect your vehicle's Anti-Lock Brake System (ABS) resulting in a serious accident.
- When replacing tires (or wheels), it is recommended to replace the two front or two rear tires (or wheels) as a pair. Replacing just one tire can seriously affect your vehicle's handling.
- Tires degrade over time, even when they are not being used. Regardless of the remaining tread, HYUNDAI recommends that tires be replaced after six (6) years of normal service.
- Heat caused by hot climates or frequent high loading conditions can accelerate the aging process. Failure to follow this warning may cause sudden tire failure, which could lead to a loss of vehicle control resulting in an accident.

Compact spare tire replacement (if equipped)

A compact spare tire has a shorter tread life than a regular size tire. Replace it when you can see the tread wear indicator bars on the tire. The replacement compact spare tire should be the same size and design tire as the one provided with your new vehicle and should be mounted on the same compact spare tire wheel. The compact spare tire is not designed to be mounted on a regular size wheel, and the compact spare tire wheel is not designed for mounting a regular size tire.



WARNING

The original tire should be repaired or replaced as soon as possible to avoid failure of the spare and loss of vehicle control resulting in an accident. The compact spare tire is for emergency use only. Do not operate your vehicle over 80 km/h (50 mph) when using the compact spare tire.

Wheel replacement

When replacing the metal wheels for any reason, make sure the new wheels are equivalent to the original factory units in diameter, rim width and offset.

Tire traction

Tire traction can be reduced if you drive on worn tires, tires that are improperly inflated or on slippery road surfaces. Tires should be replaced when tread wear indicators appear. To reduce the possibility of losing control, slow down whenever there is rain, snow or ice on the road.

Tire maintenance

In addition to proper inflation, correct wheel alignment helps to decrease tire wear. If you find a tire is worn unevenly, have your dealer check the wheel alignment.

When you have new tires installed, make sure they are balanced. This will increase vehicle ride comfort and tire life. Additionally, a tire should always be rebalanced if it is removed from the wheel.

Tire sidewall labeling

This information identifies and describes the fundamental characteristics of the tire and also provides the tire identification number (TIN) for safety standard certification. The TIN can be used to identify the tire in case of a recall.